

10th Week Summary (11/05/26)

- **ANOVA**
- ANalysis Of VAriance (ANOVA) is a popular statistical tool to analyze the effect of a categorical variable with different levels of treatments (factor) on a numerical variable (response).
- In general, we can write:

Total Variation = Variation Between Treatments + Variation Within Treatments.

• **Grand Mean:** $\bar{y}_{..} = \frac{n_1\bar{y}_1 + n_2\bar{y}_2 + \dots + n_t\bar{y}_t}{n_1 + n_2 + \dots + n_t}$

• **Total Variability:** $\sum_{i=1}^t \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2$

• **Variability Between Samples:** $\sum_{i=1}^t n_i (\bar{y}_i - \bar{y}_{..})^2$

• **Variability Within Samples:** $\sum_{i=1}^t \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2$

	Treatment Levels				
	1	2	3	...	t
y_{11}	y_{21}	y_{31}			y_{t1}
y_{12}	y_{22}	y_{32}			y_{t2}
...	...	...			...
y_{1n_1}	y_{2n_2}	y_{3n_3}			y_{tn_t}
=====					
Mean	$\bar{y}_1$	$\bar{y}_2$	$\bar{y}_3$		$\bar{y}_t$
St.dev.	s_1	s_2	s_3		s_t

df's: $\sum n_i - 1$ $t - 1$ $\sum n_i - t$

$SS(Total) = SSB + SSE$

- Hypothesis Testing

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_t$$

$$H_a : \mu_i \neq \mu_j \text{ for some pair } (i, j).$$

$$\text{Test Statistic: } F = \frac{SSB/df_B}{SSE/df_E}$$

Decision Rule: Reject H_0 in favor of H_a if $F > F_{\alpha}(df_B, df_E)$.

Source of Variation	df	Sum of Squares	Mean Square	F	p-value
Group (Between)	$t - 1$	$\sum n_i (\bar{y}_i - \bar{y}_{..})^2 = SS_B$	$\frac{SS_B}{df_B} = MS_B$	$\frac{MS_B}{MS_E} = F_{\text{calc}}$	$\Pr(F > F_{\text{calc}})$
Error (Within)	$N - t$	$\sum (n_i - 1) s_i^2 = SS_E$	$\frac{SS_E}{df_E} = MS_E$		
Total	$N - 1$	$\sum (y_{ij} - \bar{y}_{..})^2 = SS_T$			

R: `aov(Wt ~ sport + gender, data=ais2)`

- For the above ANOVA table:

$$N = \sum_i n_i$$

$$SS_T = SS_B + SS_E$$

MS_E is the pooled sample variance, an estimator for σ^2

- Assumptions:

Homogeneity of variances: $\sigma_1 = \sigma_2 = \dots = \sigma_t$.

Data are generated from normal distributions for each treatment.

- What if normality fails?

We use the non-parametric test: "Kruskal-Wallis Test"

- What if the equality of variances fails?

We "transform" the data